AMINES

Methods of formation:-

① By Reduction of Nitrogen containing functional groups.

ⓐ Nitro:-

$R-NO_2 \xrightarrow{Sn+HCl \,/\, H_2+Metal \,/\, LiAlH_4 \,/\, HI+Red\ P} R-NH_2$

$Ar-NO_2 \xrightarrow{Sn+HCl \,/\, Fe+HCl \,/\, Zn+HCl \,/\, H_2+Metal} Ar-NO_2$

$Ar-NO_2 \xrightarrow{LiAlH_4} Ar-N=N-Ar$

ⓑ Cyanide. (Medius reduction)

$R-C\equiv N \xrightarrow{H_2+Metal/\ LiAlH_4 \,/\, HI+Red.P \,/\, Na+C_2H_5OH} R-CH_2-NH_2$ 1°

(c) $R-N\equiv C \xrightarrow{\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad} R-NH-CH_3$ 2°

AMINES

Methods of formation:-

① By Reduction of Nitrogen containing functional groups.

ⓓ amide $R-\overset{\overset{O}{\|}}{C}-NH_2$ $\xrightarrow{LiAlH_4 \,/\, HI + Red.P}$ $R-CH_2-NH_2$

ⓔ oxime $R-CH=N^{OH}$ $\xrightarrow{LiAlH_4 \,/\, H_2 + Metal \,/\, Na + C_2H_5OH}$ $R-CH_2-NH_2$

ⓕ Imine $R-\underset{\underset{H}{|}}{C}=N-R'$ $\xrightarrow[HI + Red\ P]{" \,/\, " \,/\, " \ "}$ $R-CH_2-NH-R'$

ⓖ azide $R-N_3$ $\xrightarrow{LiAlH_4 \,/\, NaBH_4}$ $R-NH_2 + N_2$

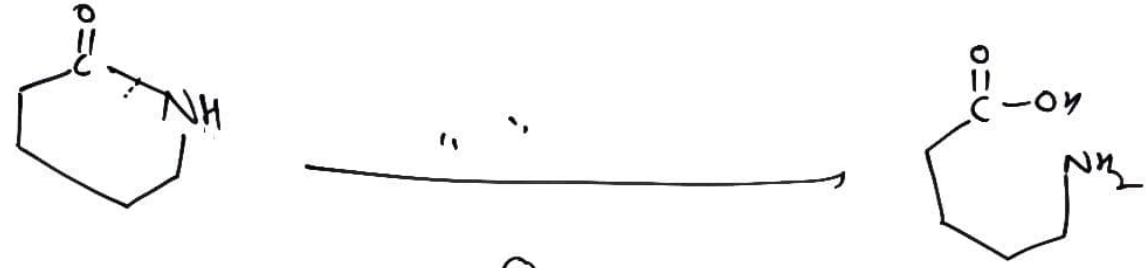
ⓗ isocyanate $R-N=C=O$ $\xrightarrow{LiAlH_4 \,/\, HI + Red\,P}$ $R-NH-CH_3$

AMINES

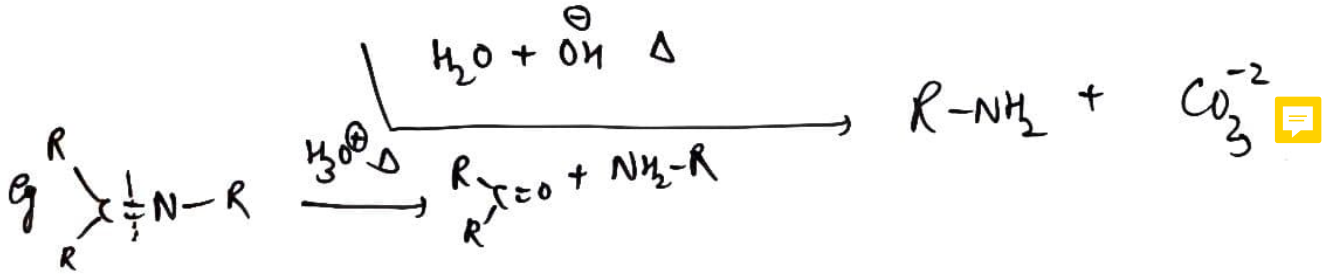
Methods of formation:-

② By Hydrolysis of $R-\overset{O}{\overset{\|}{C}}-NH-R$ / $R-N=C=O$ (isocyanate) / $>C=N-$ (imine)

Eg $R-\overset{O}{\overset{\|}{C}}-NH-R' \xrightarrow{H_2O + H^{\oplus}/OH^{\ominus} + \Delta} R-\overset{O}{\overset{\|}{C}}-OH + H_2N-R'$

Eg 

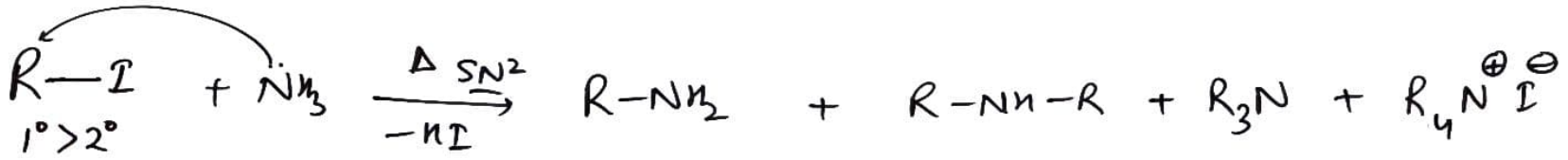
Eg $R-N=C=O \xrightarrow{H_2O + H^{\oplus}\ \Delta} R-NH_2 + CO_2$

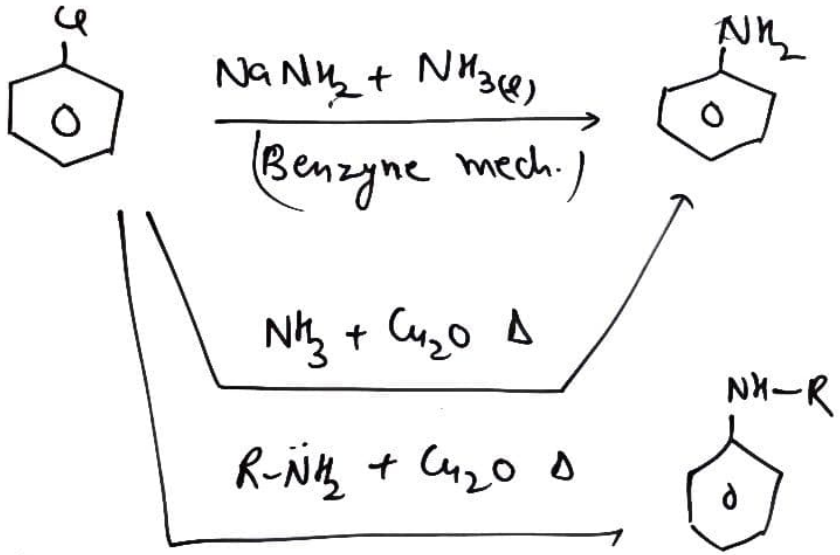


AMINES

Methods of formation:-

(3) By Hafmann's ammonolysis of alkyl halide:-

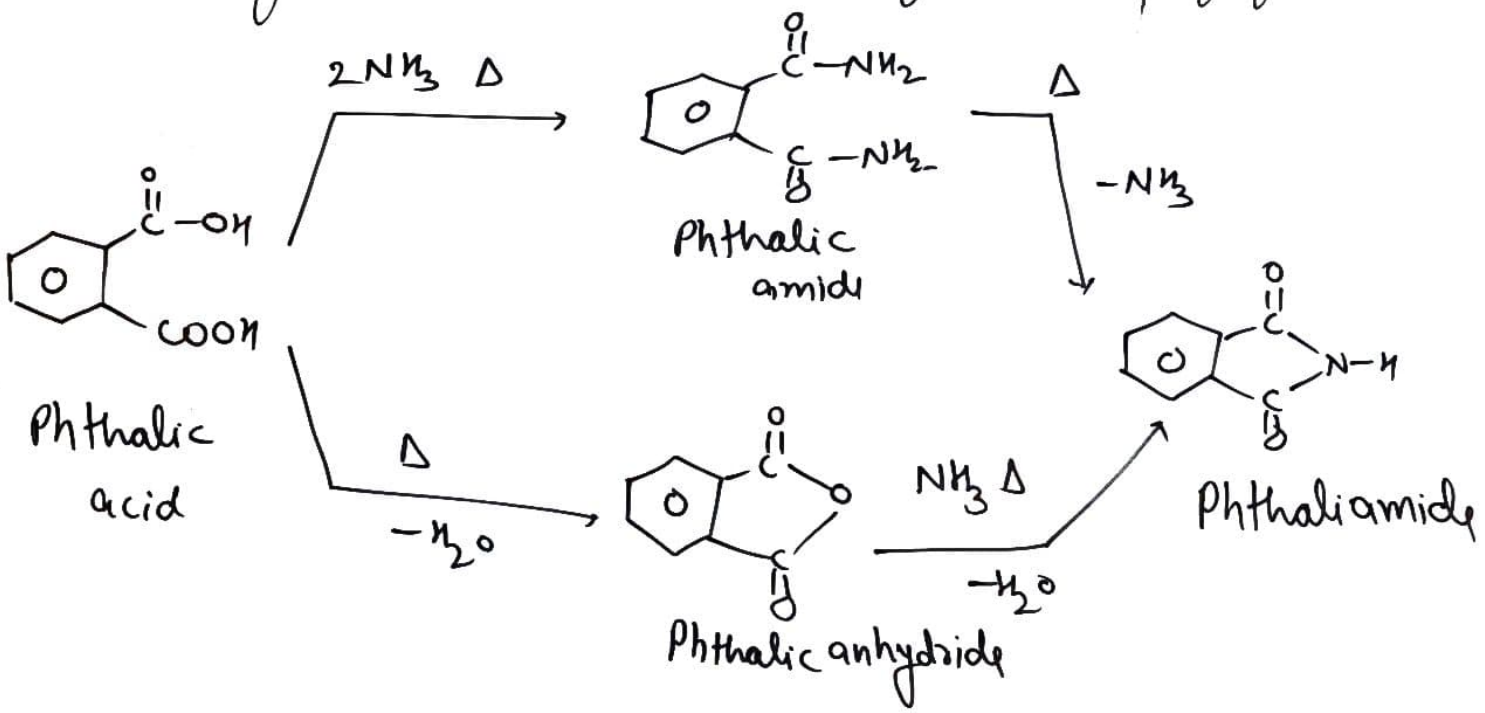




AMINES

Methods of Formation:-

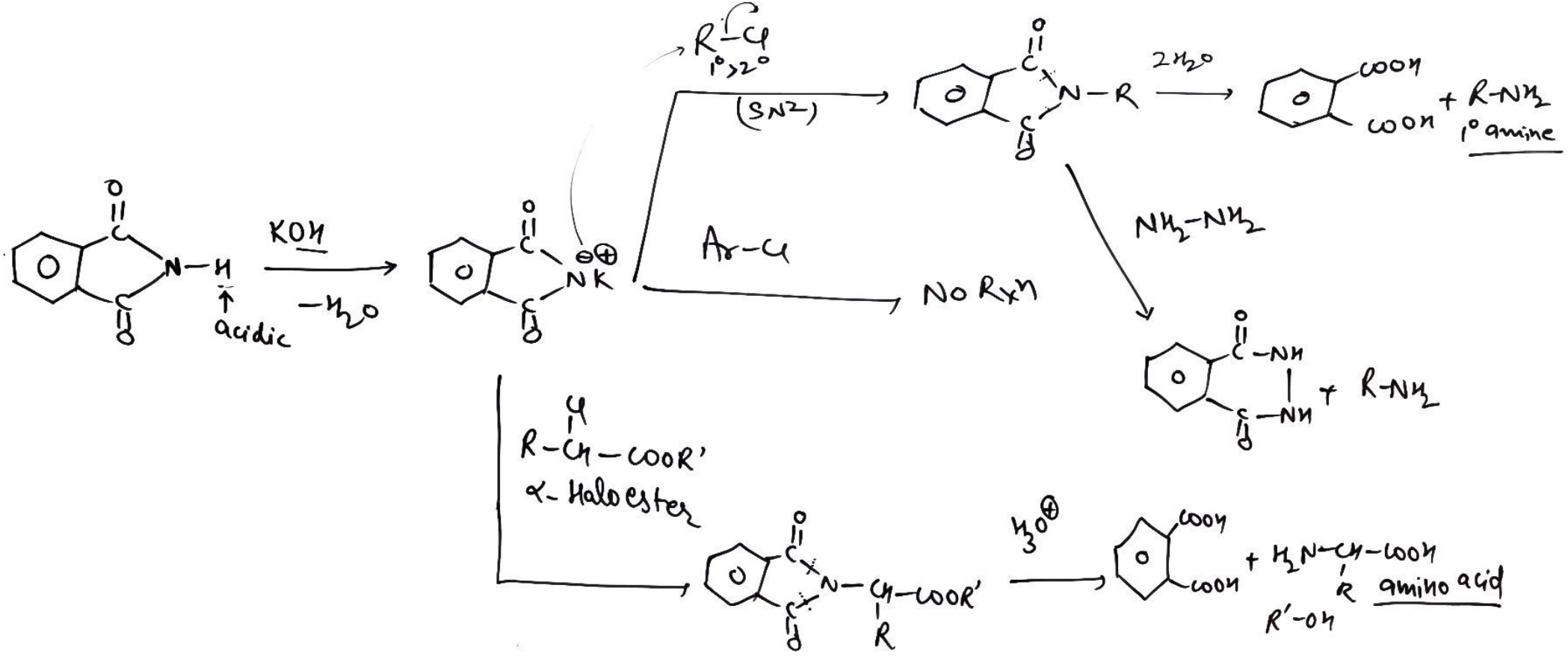
(4) By Gabriel – Phthalimide synthesis (only for 1° aliphatic amine)



AMINES

Methods of Formation:-

(4) By Gabriel – Phthalimide synthesis (only for 1° aliphatic amine)



AMINES

Methods of formation:-

(5) By Grignard Reagent:-

$$R{-}MgCl + Cl{-}NH_2 \longrightarrow R{-}NH_2 + MgCl_2$$

Chloramine

$$\text{Cyclohexyl-MgCl} + Cl{-}NH_2 \longrightarrow \text{Cyclohexyl-}NH_2 + \text{''}$$

$$Ph{-}MgCl + Cl{-}NH_2 \longrightarrow Ph{-}NH_2 + \text{''}$$

$$\text{Cyclohexyl-MgCl} + Cl{-}NH{-}CH_3 \longrightarrow \text{Cyclohexyl-}NH{-}CH_3 + \text{''}$$

$$R{-}MgCl + Cl{-}O{-}CH_3 \longrightarrow R{-}O{-}CH_3 + MgCl_2$$

AMINES

Methods of formation:-

By Rearrangement of e⁻ deficient N containing intermediate

(6) Hofmann's Bromamide degradation (Rearrangement)

or KOBr

intramolecular rearrangement

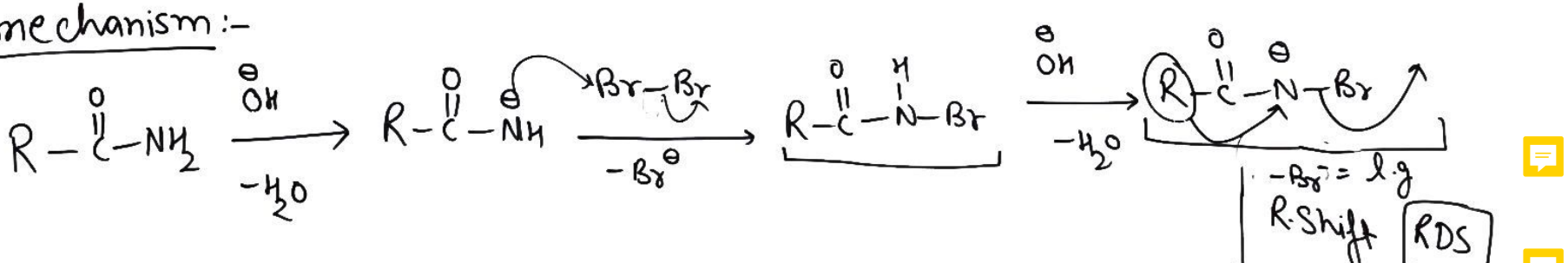
$(X_2 + \overset{\ominus}{O}H)$ or $\overset{\ominus}{O}X$ Hypohalite

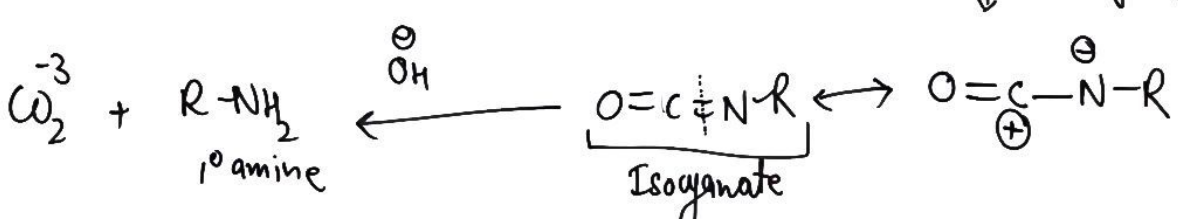


1° amide only

1° amine

mechanism:-





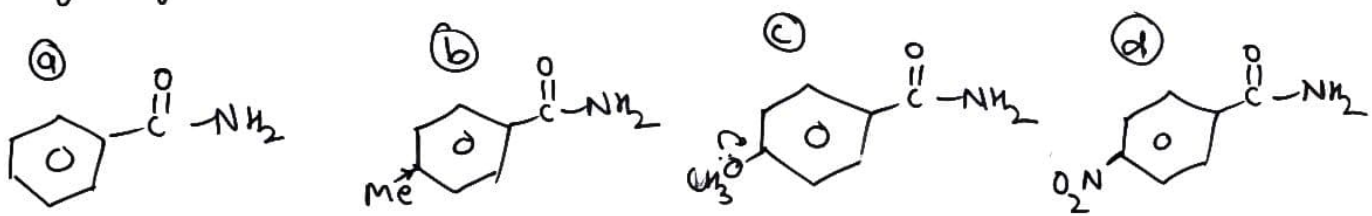
AMINES

Methods of formation:-

By Rearrangement of e⁻ deficient N containing intermediate

(6) Hofmann's Bromamide degradation (Rearrangement)

Q Rate of Hofmann Rxⁿ?



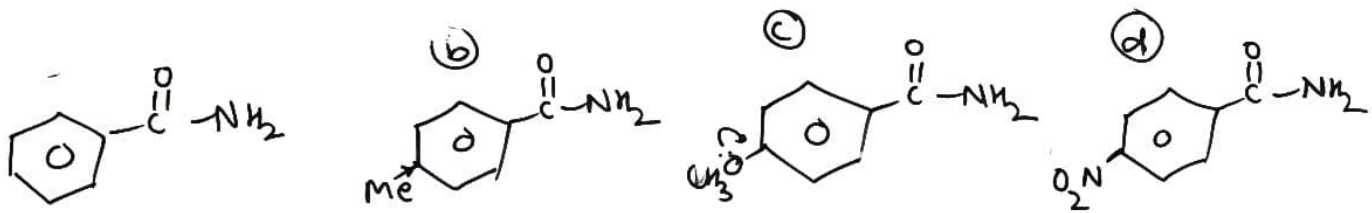
Reactivity = c > b > a > d

AMINES

Methods of formation:-

By Rearrangement of e⁻ deficient N containing intermediate

(6) Hofmann's Bromamide degradation (Rearrangement)



Reactivity = c > b > a > d

eg1 $\left[Me-\overset{O}{\overset{\|}{C}}-\overset{14}{N}H_2 + Et-\overset{O}{\overset{\|}{C}}-\overset{15}{N}H_2 \right] \xrightarrow[\text{or } Br_2 + KOH]{KOBr} Me-\overset{14}{N}H_2 + Et-\overset{15}{N}H_2$

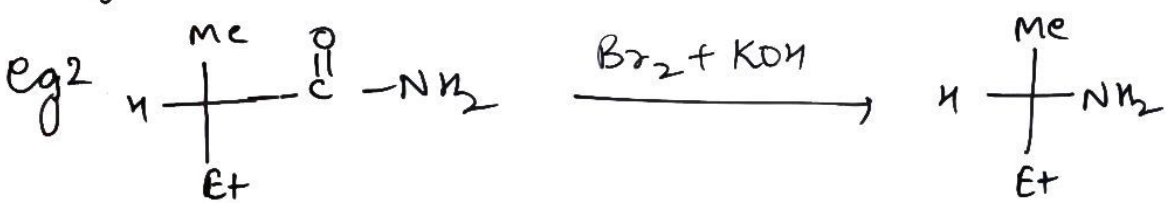
intramolecular rex.
so no mixed products

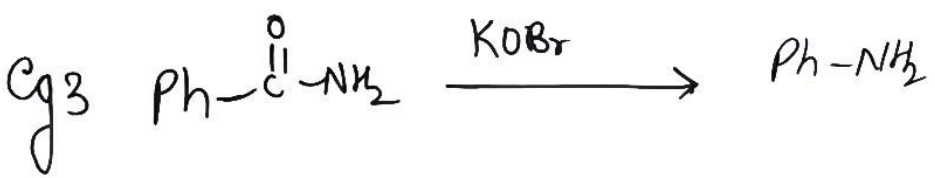
AMINES

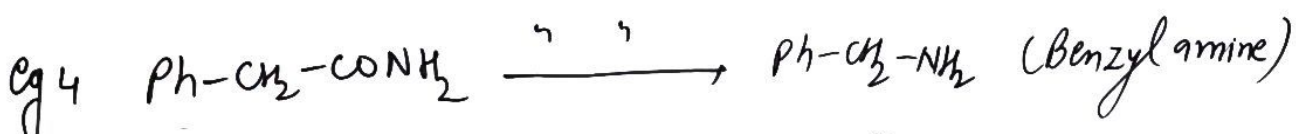
Methods of formation:-

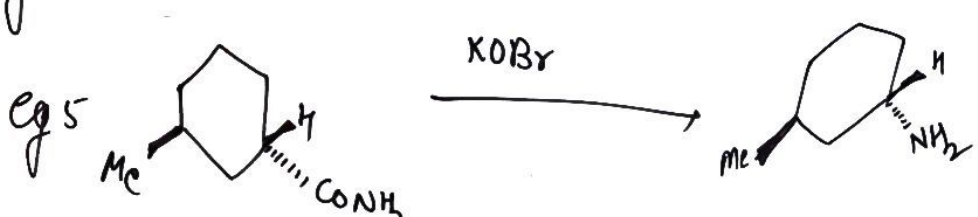
By Rearrangement of e⁻ deficient N containing intermediate

(6) Hofmann's Bromamide degradation (Rearrangement)









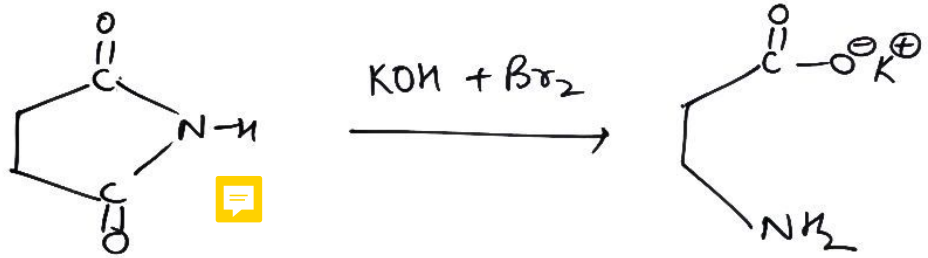
AMINES

Methods of formation:-

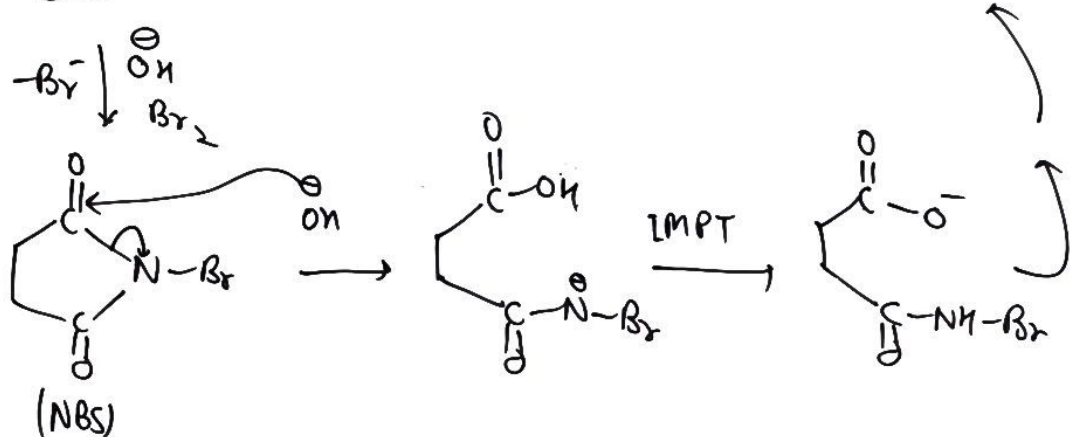
By Rearrangement of e⁻ deficient N containing intermediate

(6) Hofmann's Bromamide degradation (Rearrangement)

eg 6



Succinamide



(NBS)

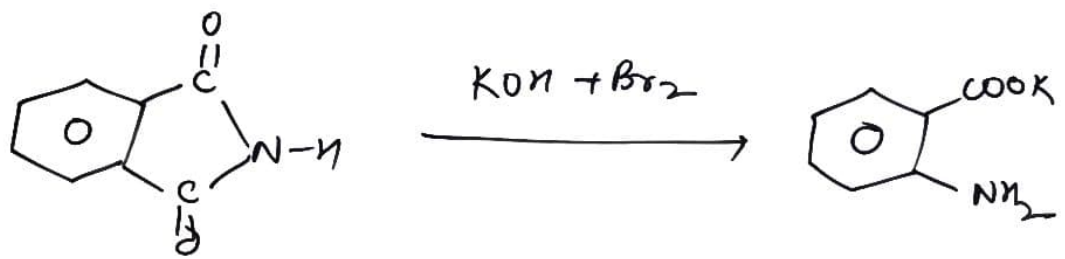
AMINES

Methods of formation:-

By Rearrangement of e^- deficient N containing intermediate

(6) Hofmann's Bromamide degradation (Rearrangement)

eg:

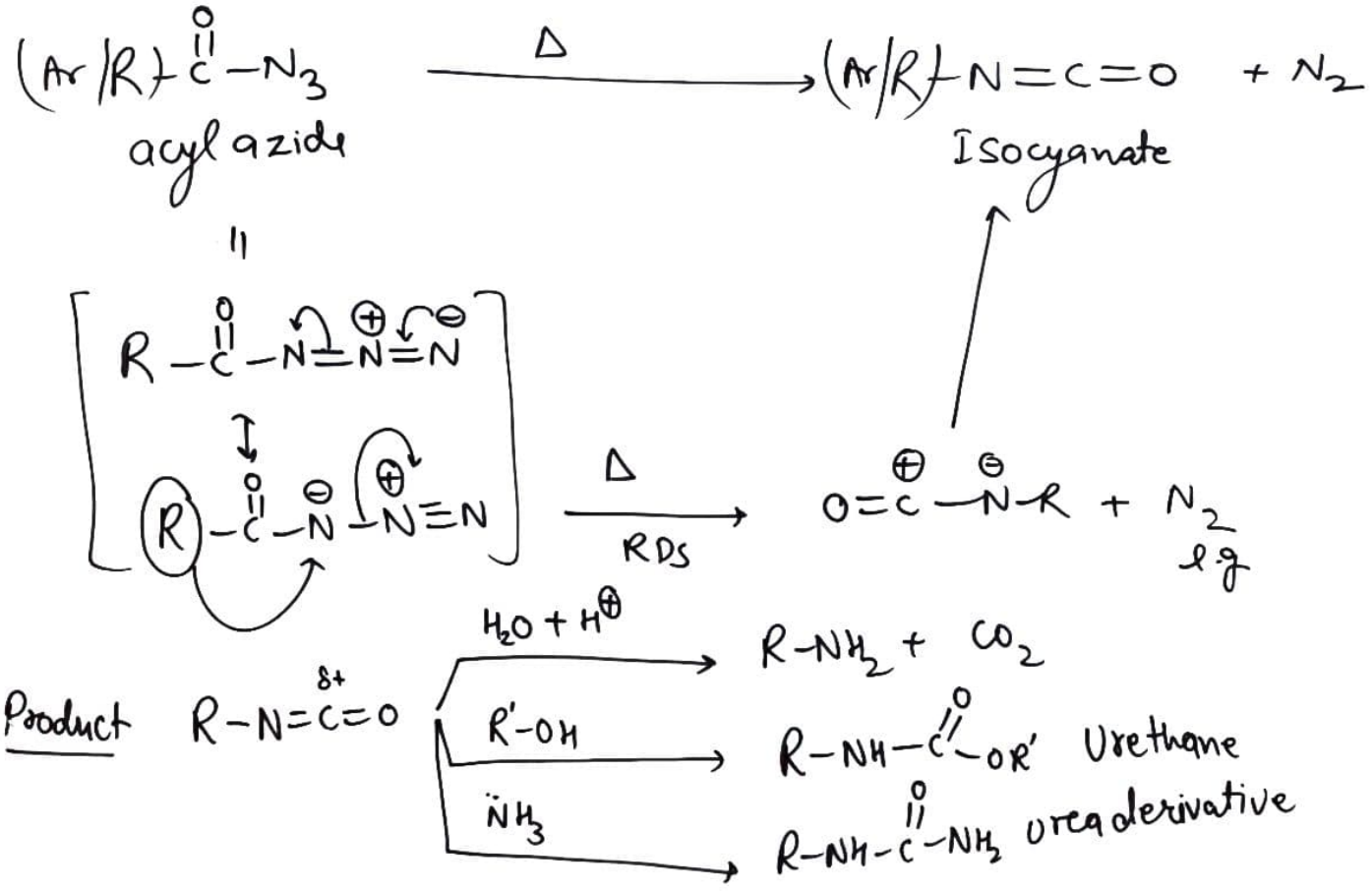


Phthalimide

AMINES

Methods of Formation:-

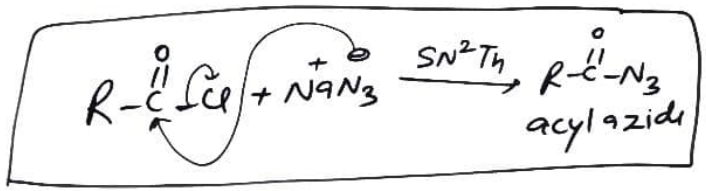
(7) By Curtius Reaction :-



AMINES

Methods of formation:-

⑧ By Schimdt Reaction:-



$$R-\overset{O}{\overset{\|}{C}}-OH + H-N_3 \xrightarrow[\Delta]{\text{conc. } H_2SO_4} R-NH_2 + CO_2 + N_2$$

Hydrazoic acid

1° amine

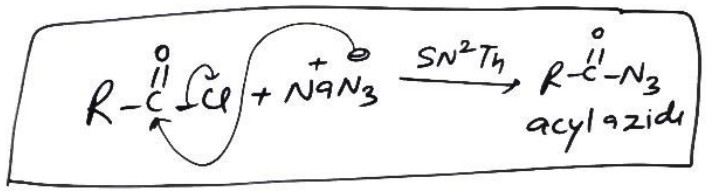
$$R-\overset{O}{\overset{\|}{C}}-N_3 \xrightarrow[-N_2]{\Delta} R-N=C=O$$

eg $Ph-COOH + HN_3 \xrightarrow{H_2SO_4} Ph-NH_2$

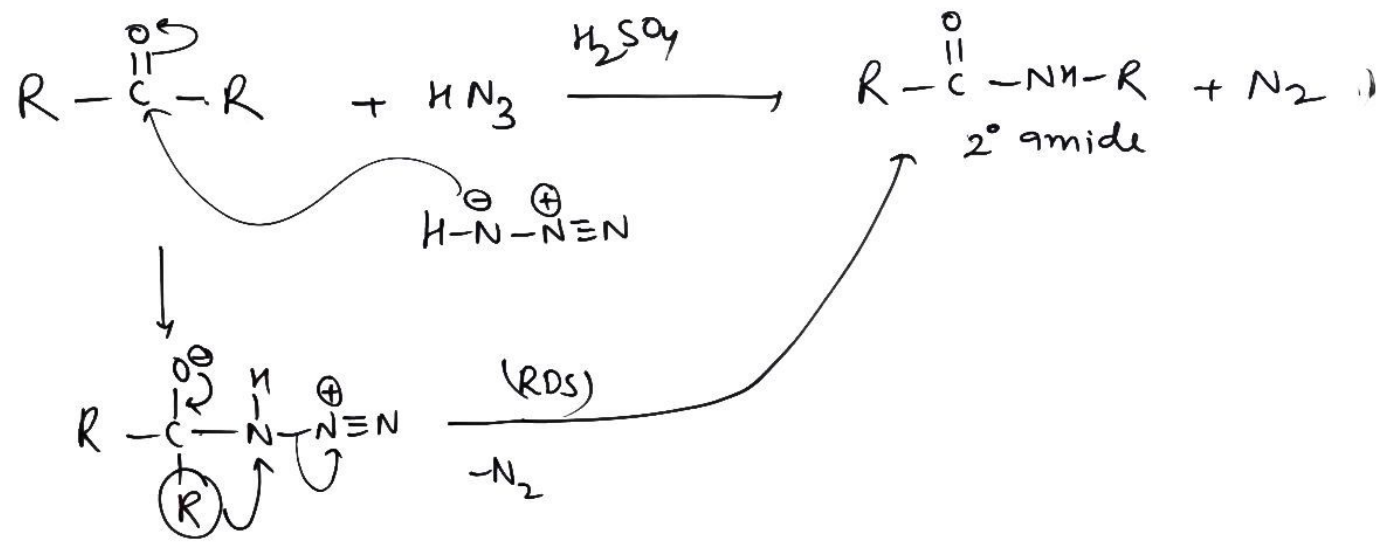
AMINES

Methods of formation:-

⑧ By Schimdt Reaction:-



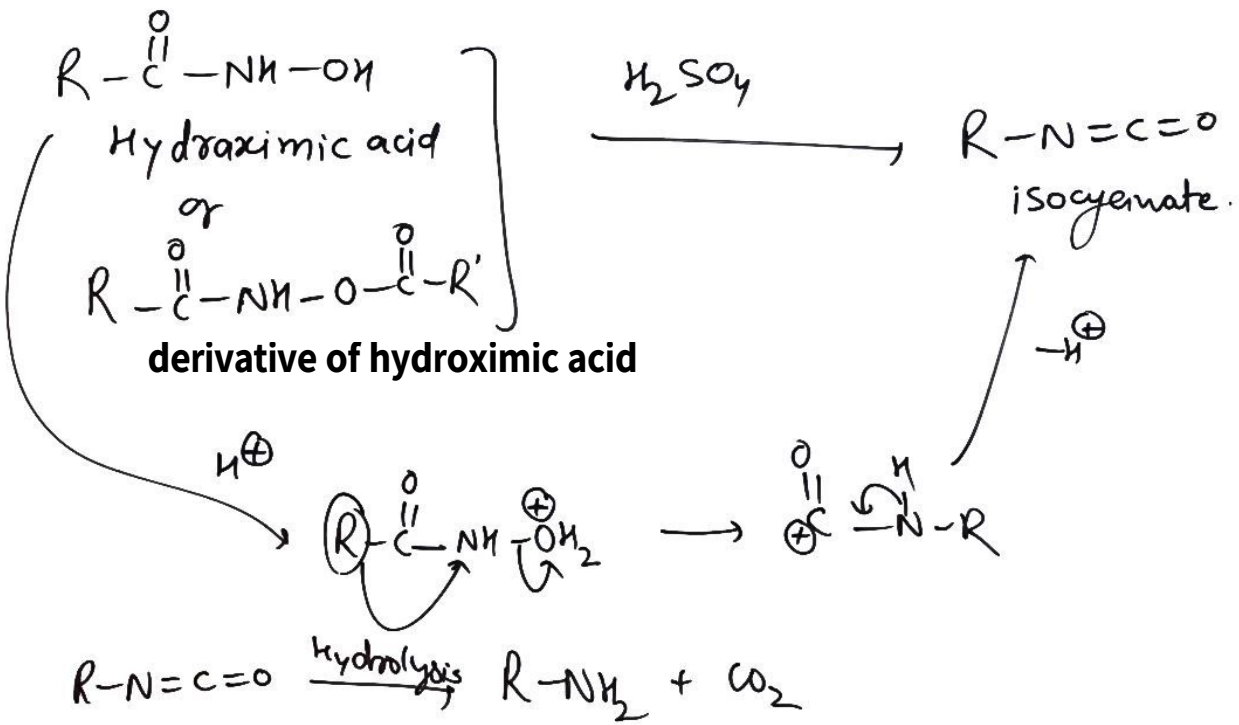
Schimdt Rxⁿ in Ketones



AMINES

Methods of formation:-

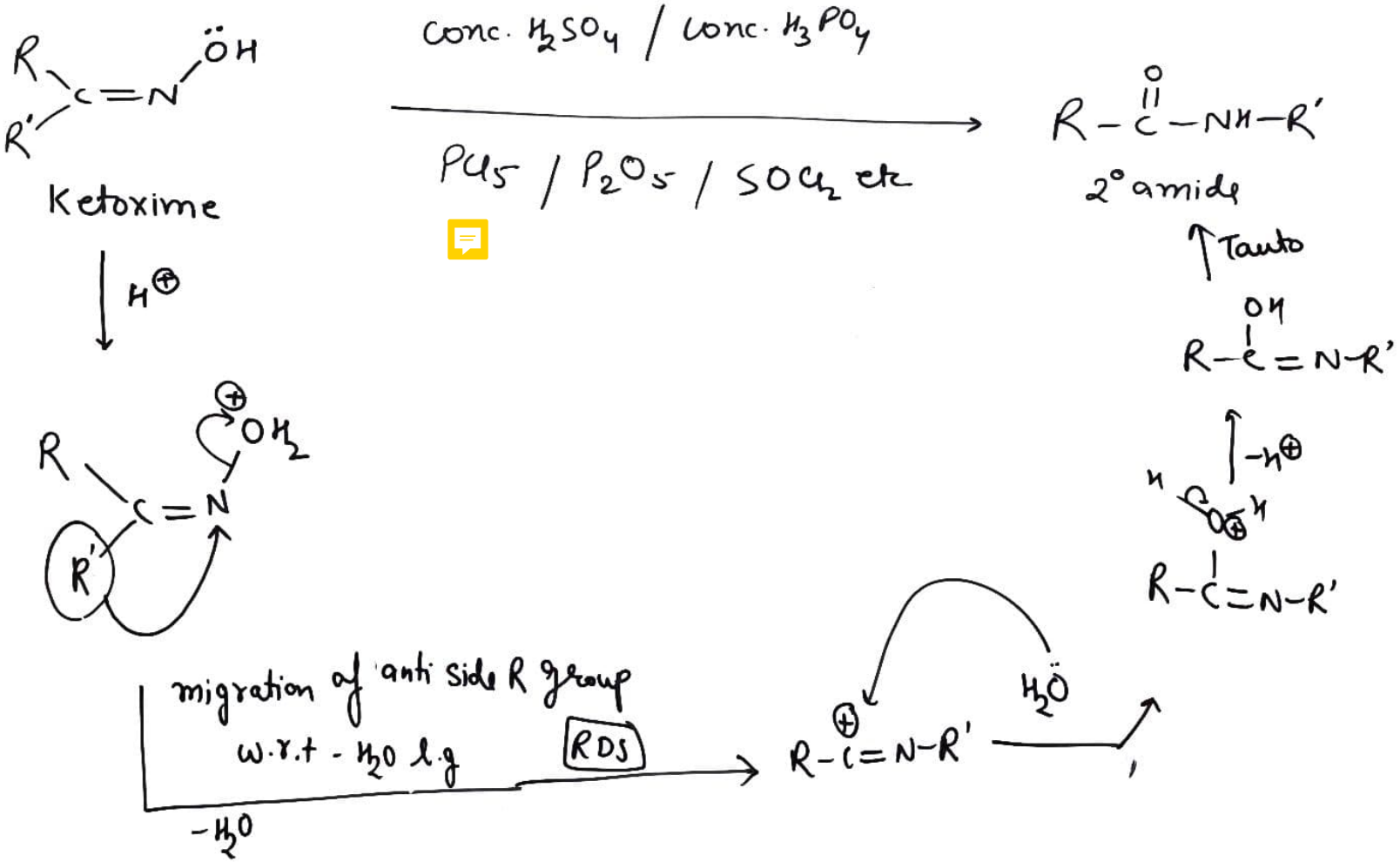
(9) Lossen Rearrangement:-



$$R-N=C=O \xrightarrow{\text{Hydrolysis}} R-NH_2 + CO_2$$

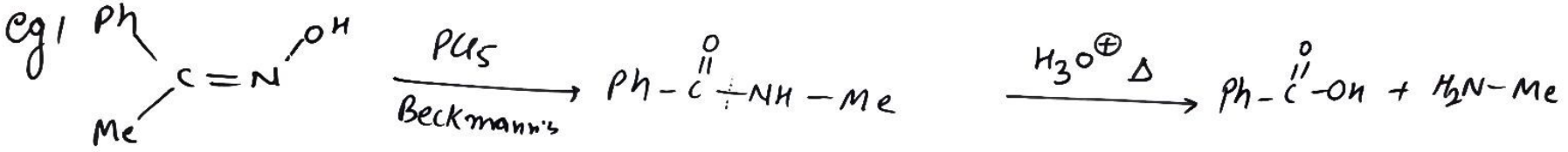
AMINES

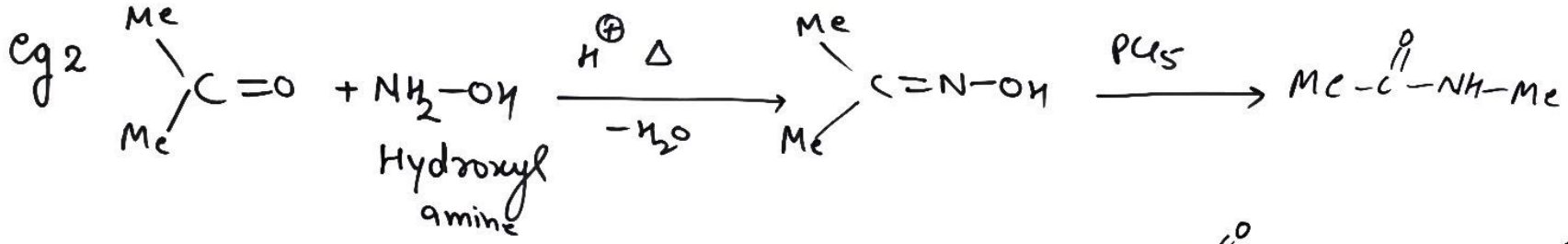
Beckmann's Rearrangement:-

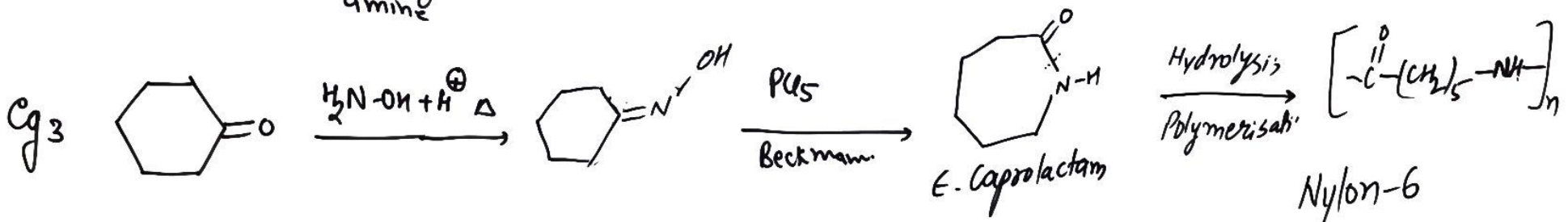


AMINES

Beckmann's Rearrangement:-

Eg 1 

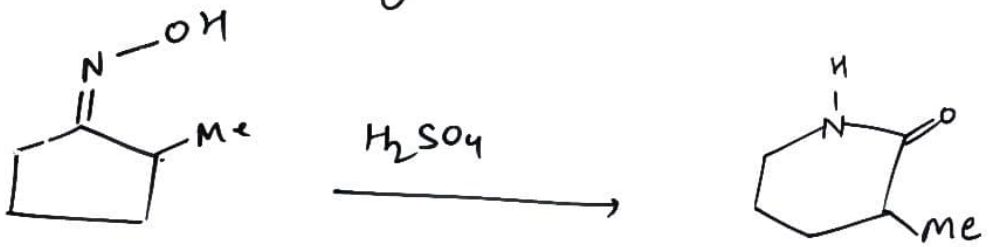
Eg 2 

Eg 3 

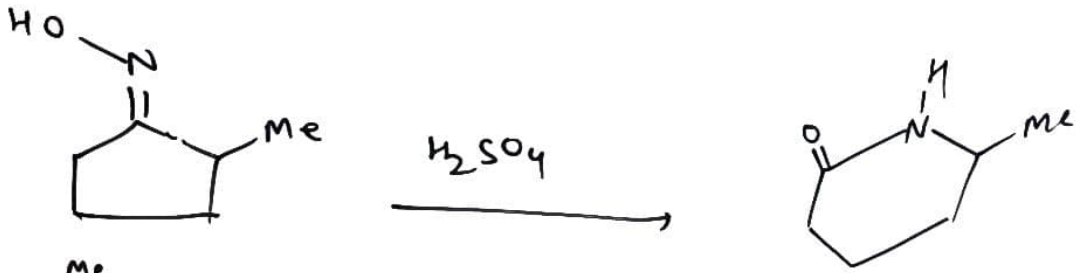
AMINES

Beckmann's Rearrangement:-

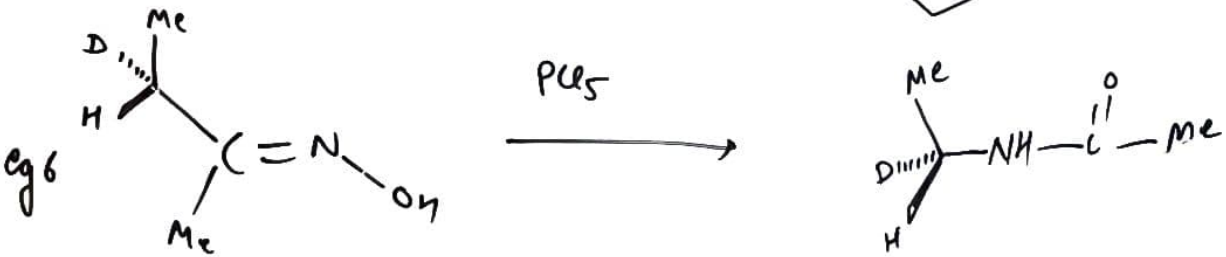
eg 4



eg 5



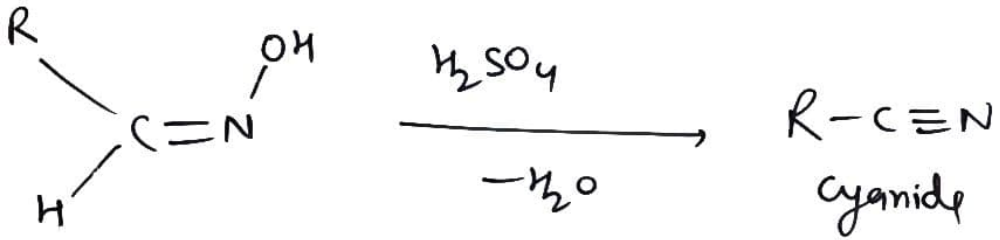
eg 6



AMINES

Beckmann's Rearrangement:-

Note for Aldoximes

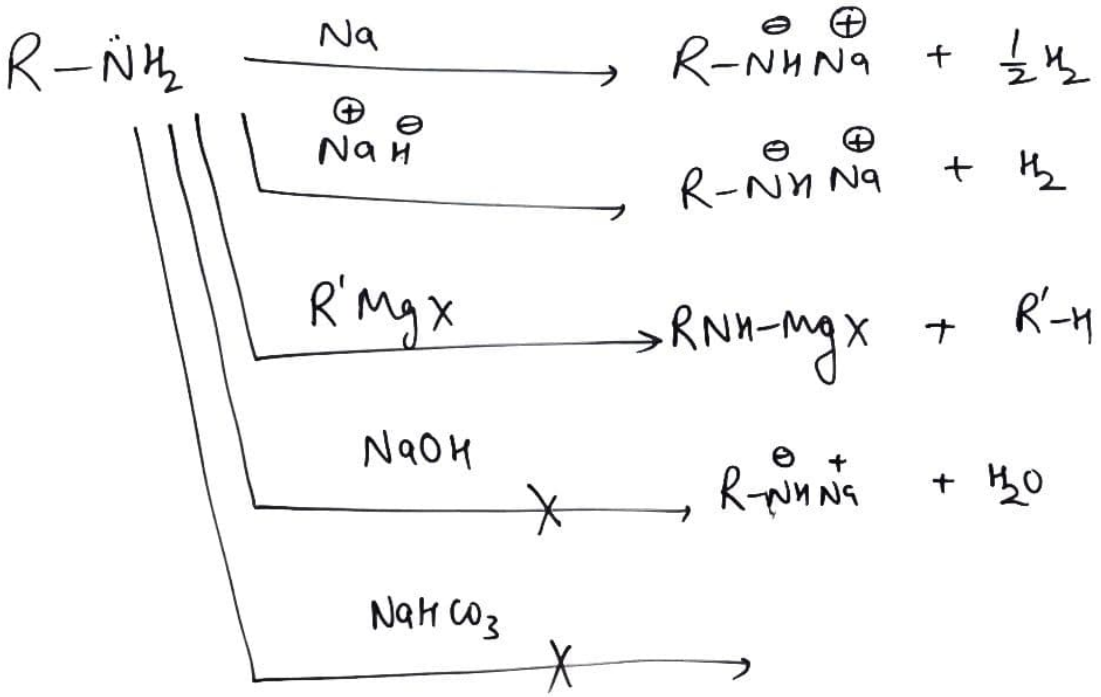


AMINES

Chemical Reactions of amines:

all reactions from hereon will be given by both aliphatic and aromatic amines unless specified otherwise

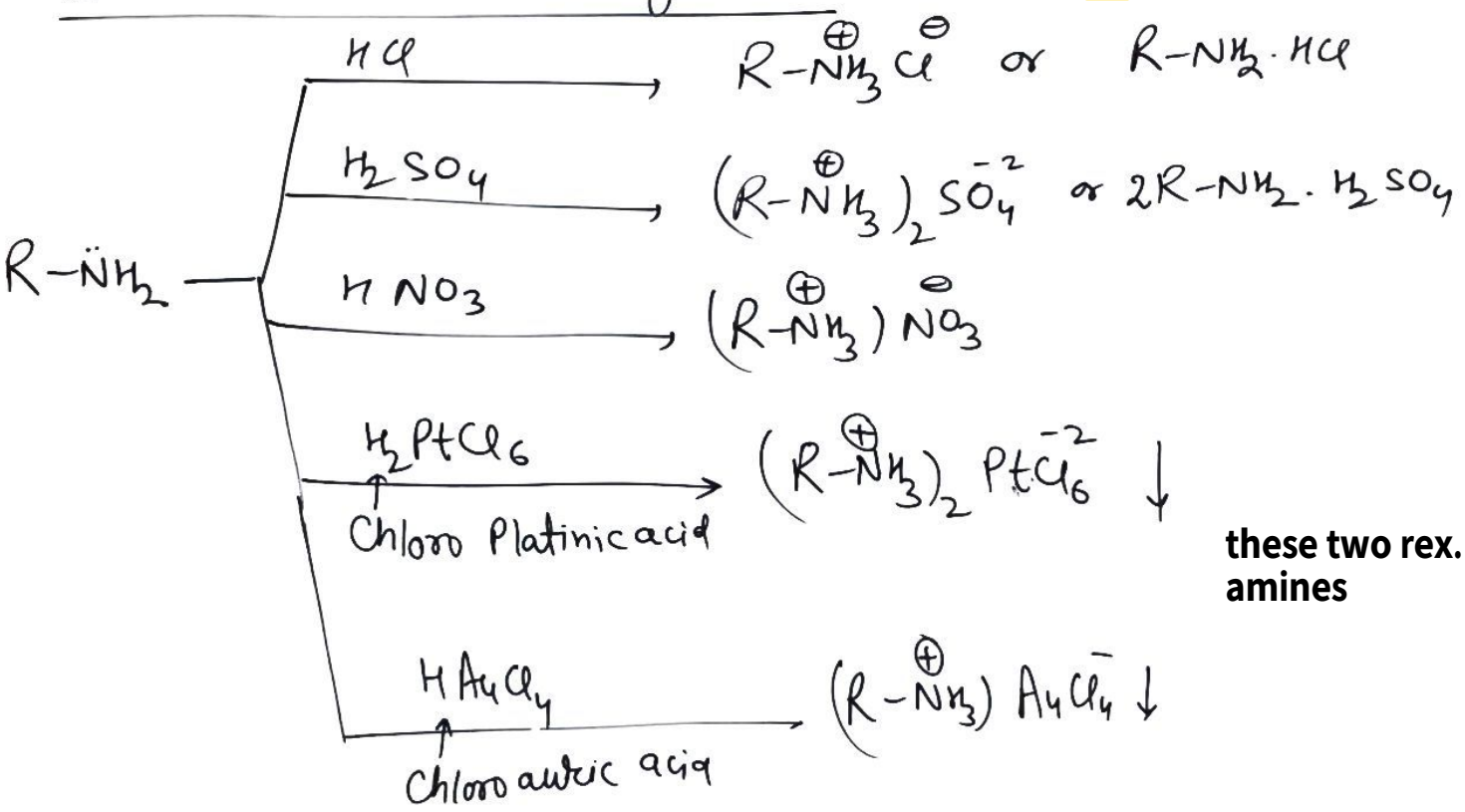
① Reaction with Strong base:-



AMINES

Chemical Reactions of amines:-

② Reaction with Strong acids:-

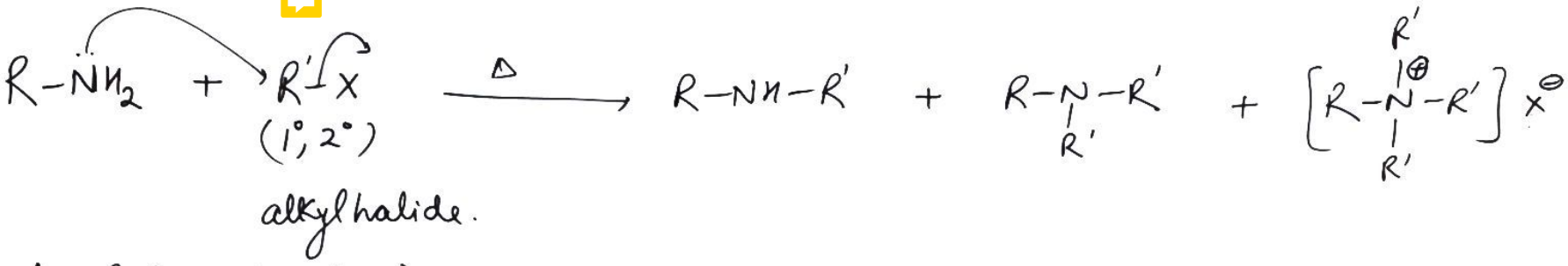


these two rex. are used for the isolation of amines

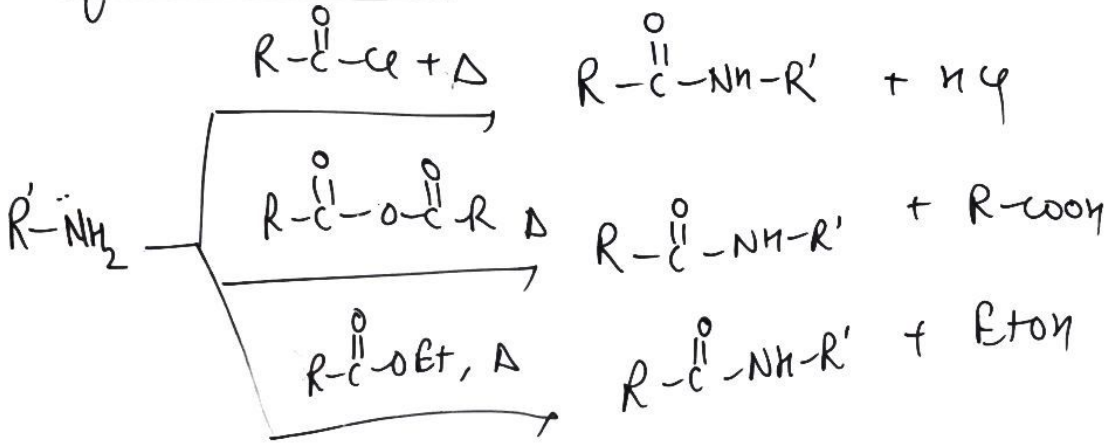
AMINES

Chemical Reactions of amines:-

③ Alkylation :-



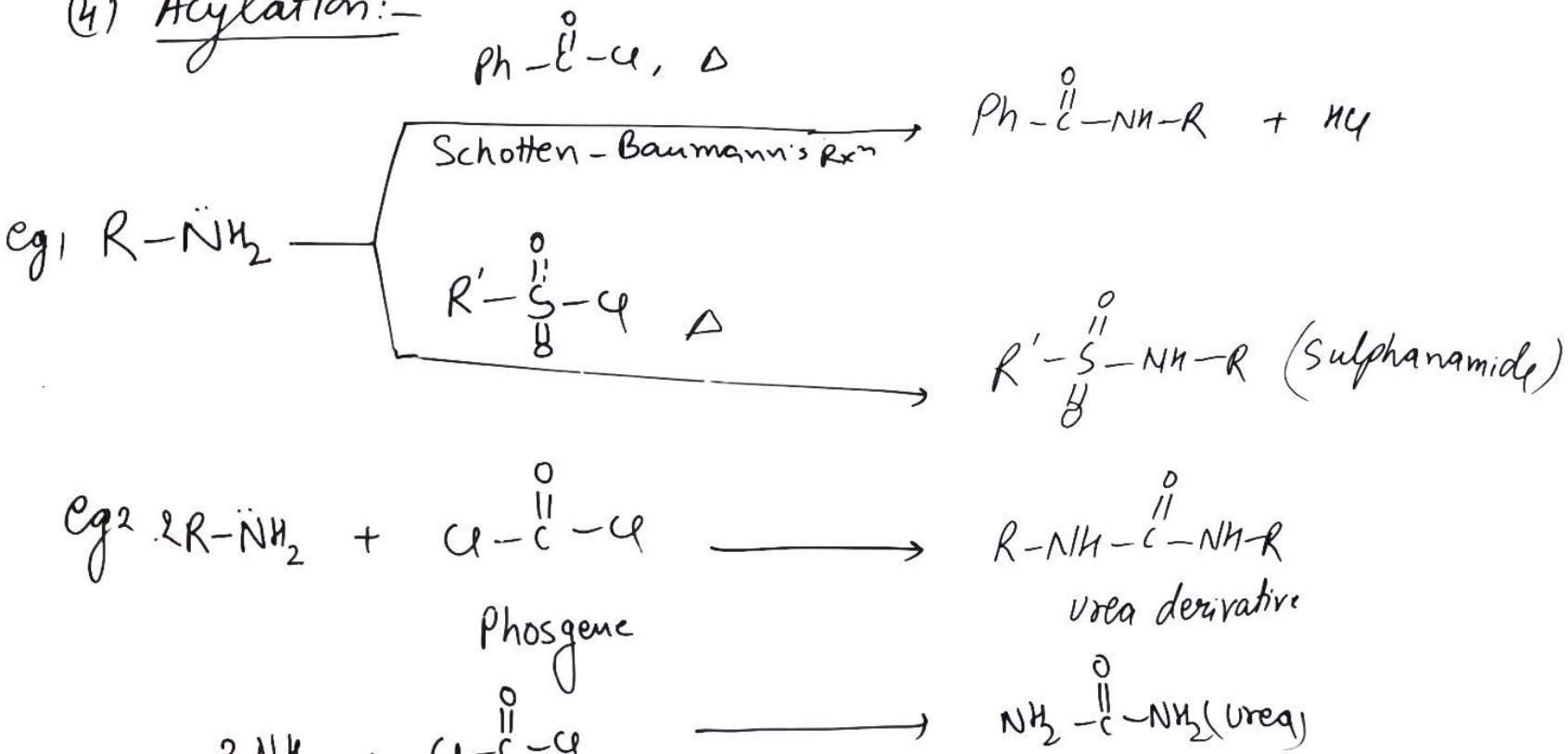
④ Acylation (SN^2 Th)



AMINES

Chemical Reactions of amines:-

(4) Acylation:-



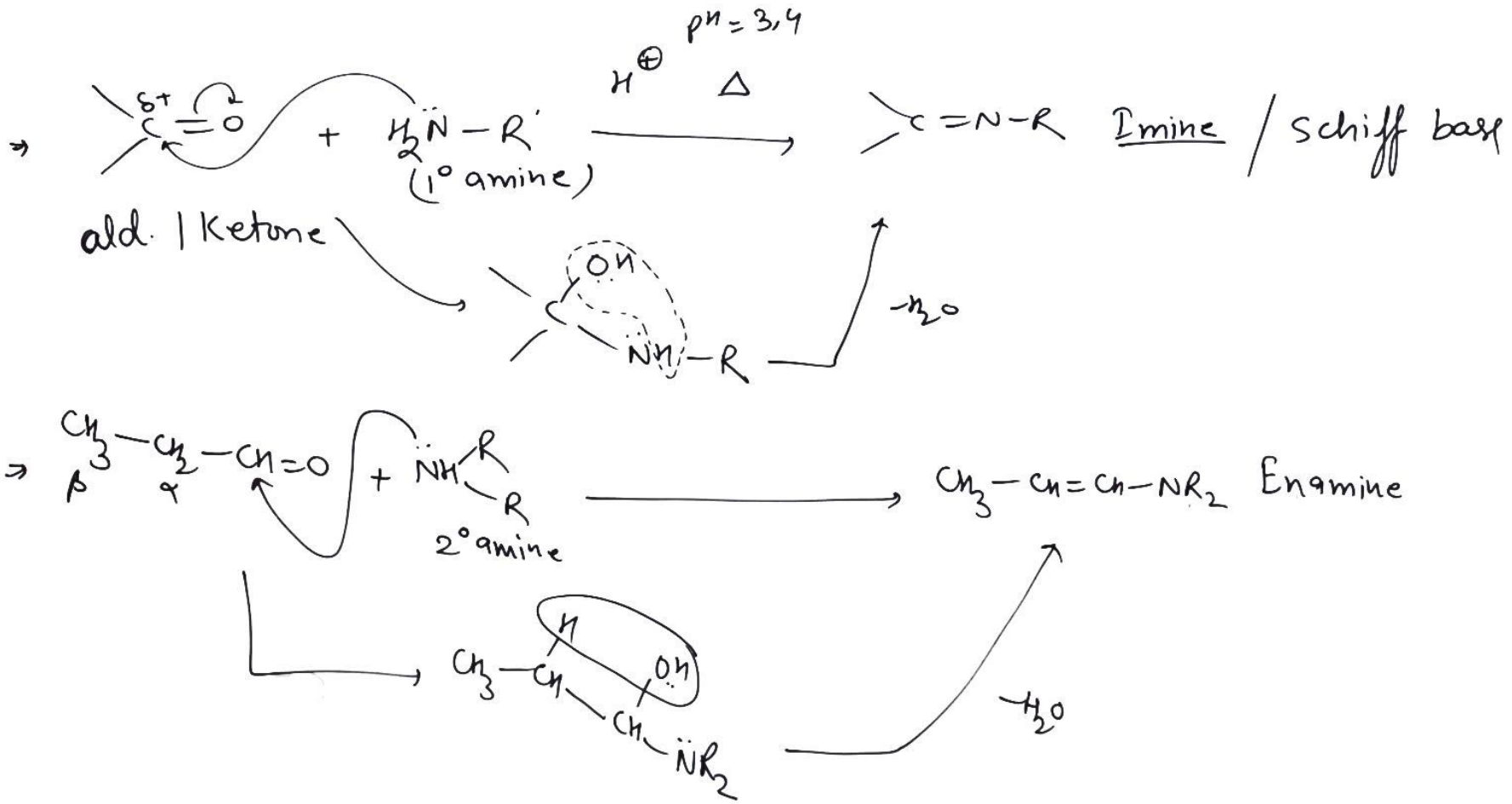
Phosgene



AMINES

Chemical Reactions of amines:-

(5) Reaction with carbonyl comp.



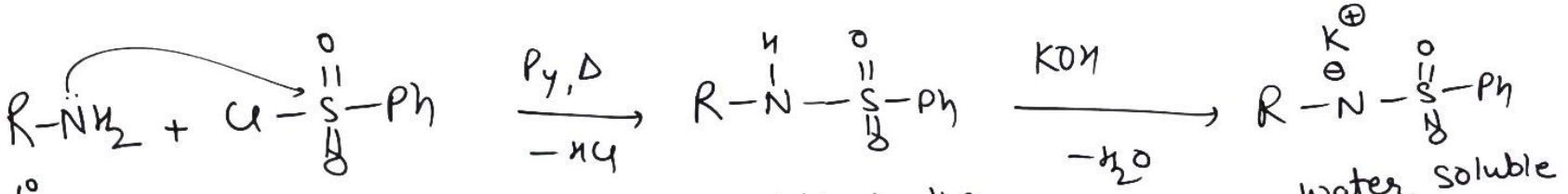
AMINES

Chemical Reactions of amines:-

(6) Reaction with Hinsberg's Reagent ($Ph-SO_2-Cl$)

↑ Benzene sulphonyl chloride

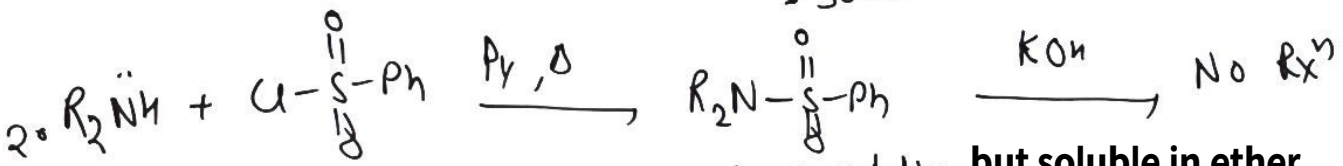
⇒ Test for 1°, 2° amine



1°

- insoluble in H_2O
- soluble in ether

water soluble



water insoluble **but soluble in ether**

3° R_3N + $Cl-SO_2-Ph$ ⟶ No Rxn

⇒ Separation of 1°, 2°, 3° amine

: in a mixture of amines, add Hisnberg reagent and filter the ppt. The ppt will contain primary and secondary amine and the filtrate will contain tertiary amine. Now, to the ppt. add KOH and water; the salt of primary amine will dissolve in water and will hence be separated

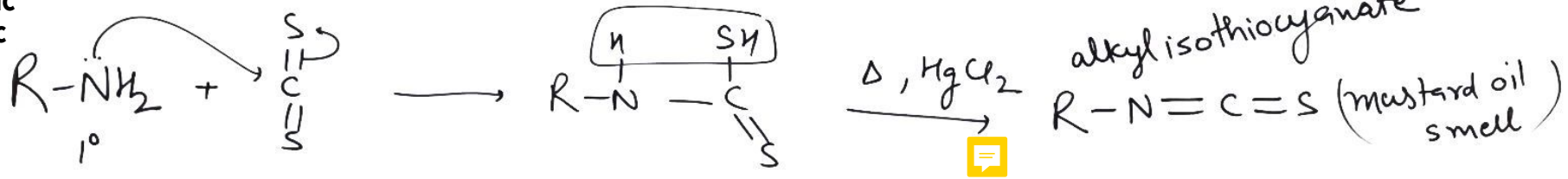
AMINES

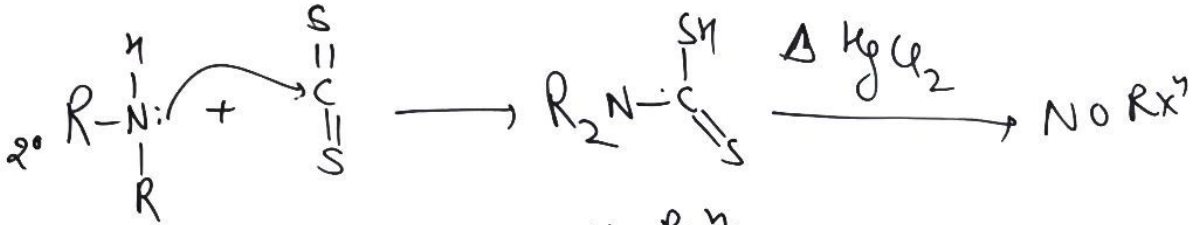
Chemical Reactions of amines:-

⑦ Reaction with CS_2

Hafmann's Mustard oil test (Test for 1° amine)

both aliphatic and aromatic amines





3° $R_3N + CS_2 \longrightarrow$ NO Rxn.

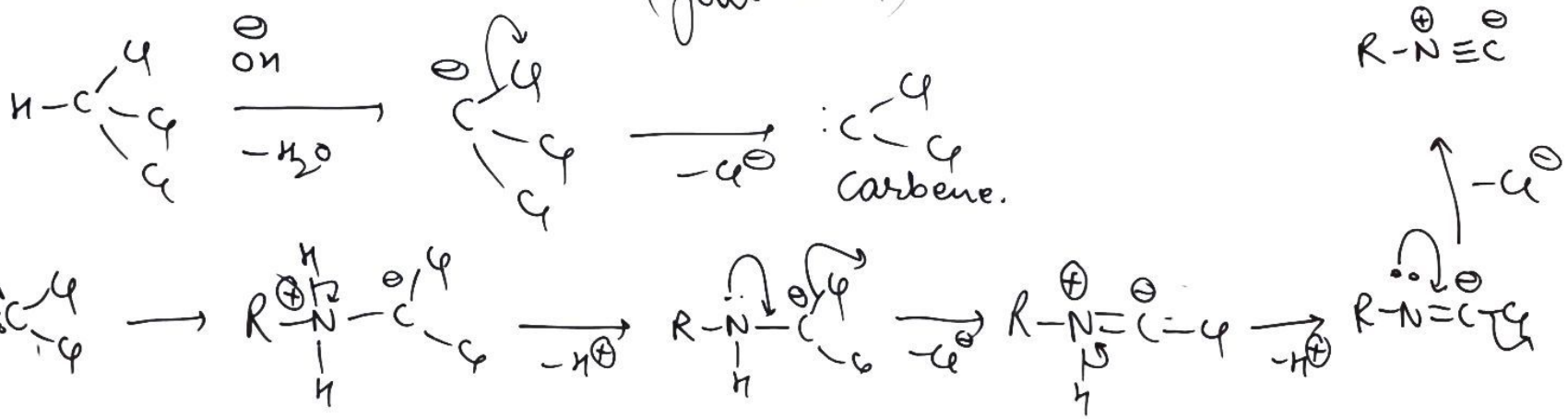
AMINES

Chemical Reactions of amines:-

⑧ Isocyanide test / Carbylamine test (Test of 1° amine)



mech.

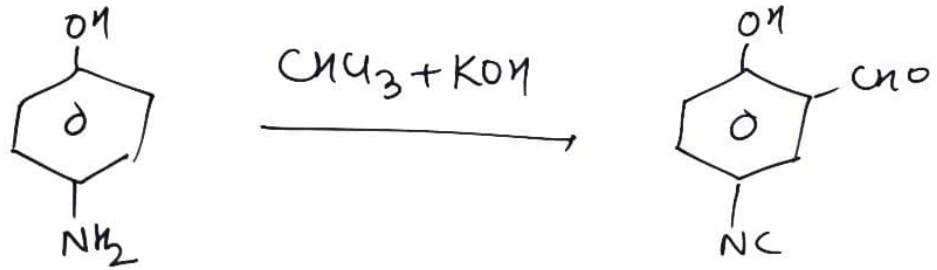


AMINES

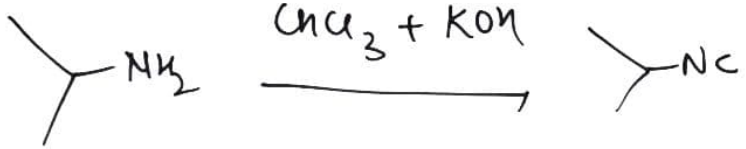
Chemical Reactions of amines:-

⑧ Isocyanide test / Carbylamine test (Test of 1° amine)

eg1



eg2

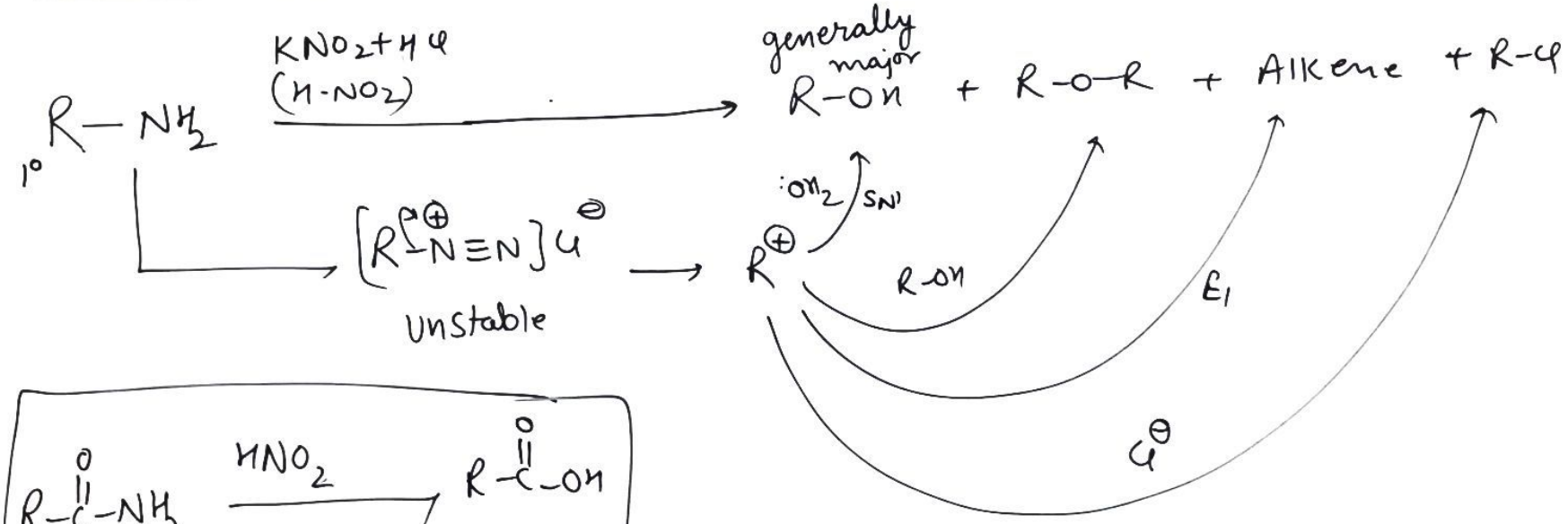


AMINES

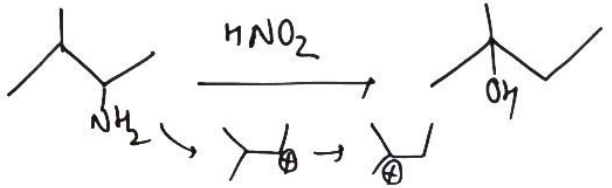
Chemical Reactions of amines:-

(9) Reaction with nitrous acid:-

⇒ with aliphatic 1° amine



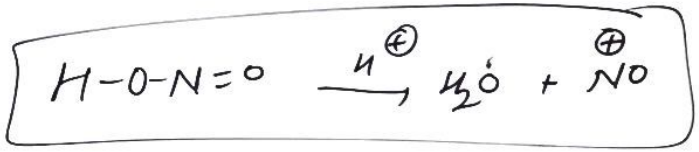


⇒ 

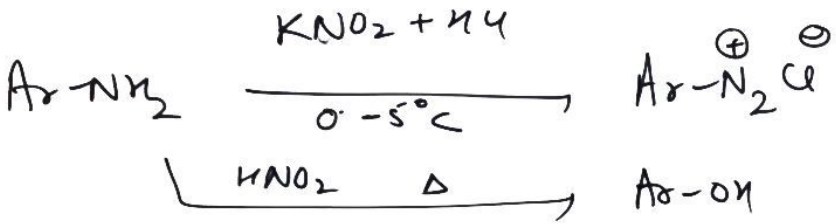
AMINES

Chemical Reactions of amines:-

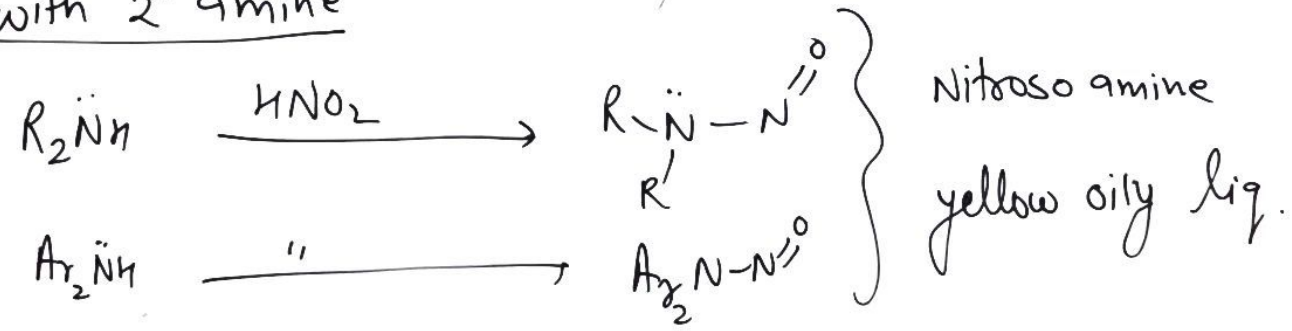
(a) Reaction with nitrous acid:-



(b) with aromatic 1° amine:-



(c) with 2° amine

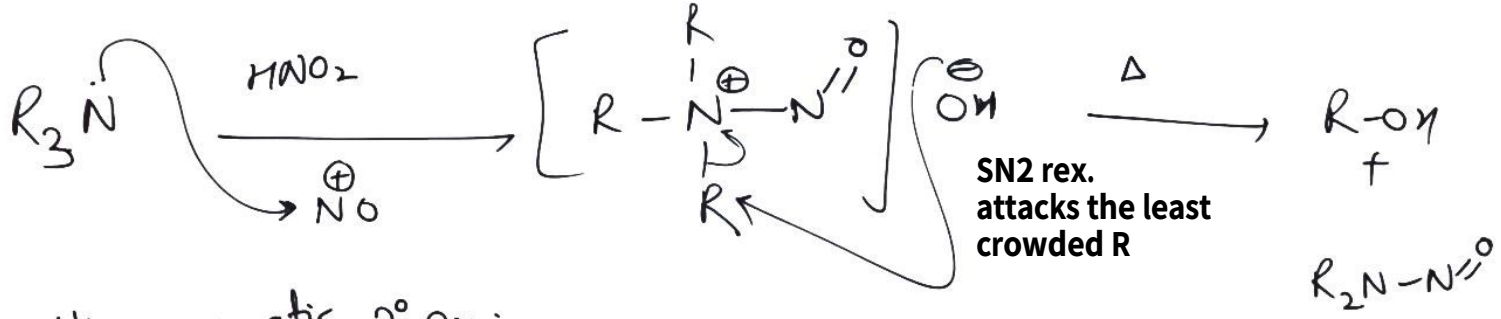


AMINES

Chemical Reactions of amines:-

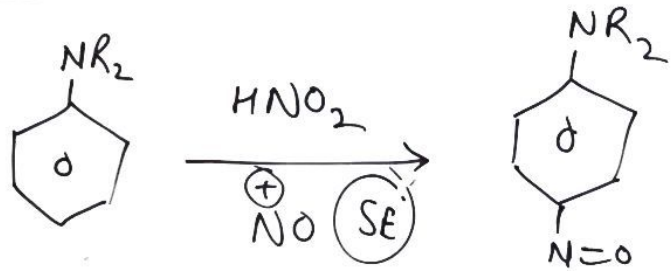
(9) Reaction with nitrous acid:-

(d) with aliphatic 3° amine:-



SN2 rex. attacks the least crowded R

(e) with aromatic 3° amine:-

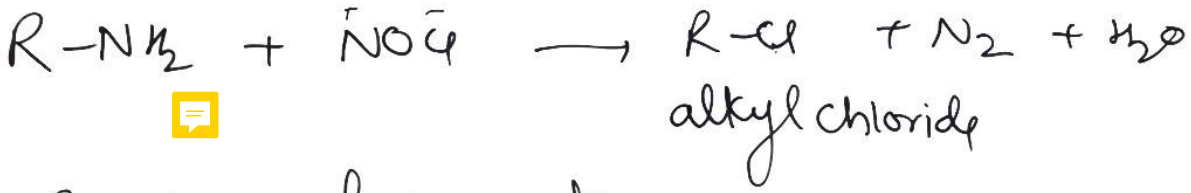


AMINES

Chemical Reactions of amines:-

(10) Reaction with Nitrosyl chloride (NOCl) Tyldane Reagent

→ (Reaction of aliphatic 1° amine only)



(11) Reaction of aromatic amine:-

(11.A) Diazo coupling

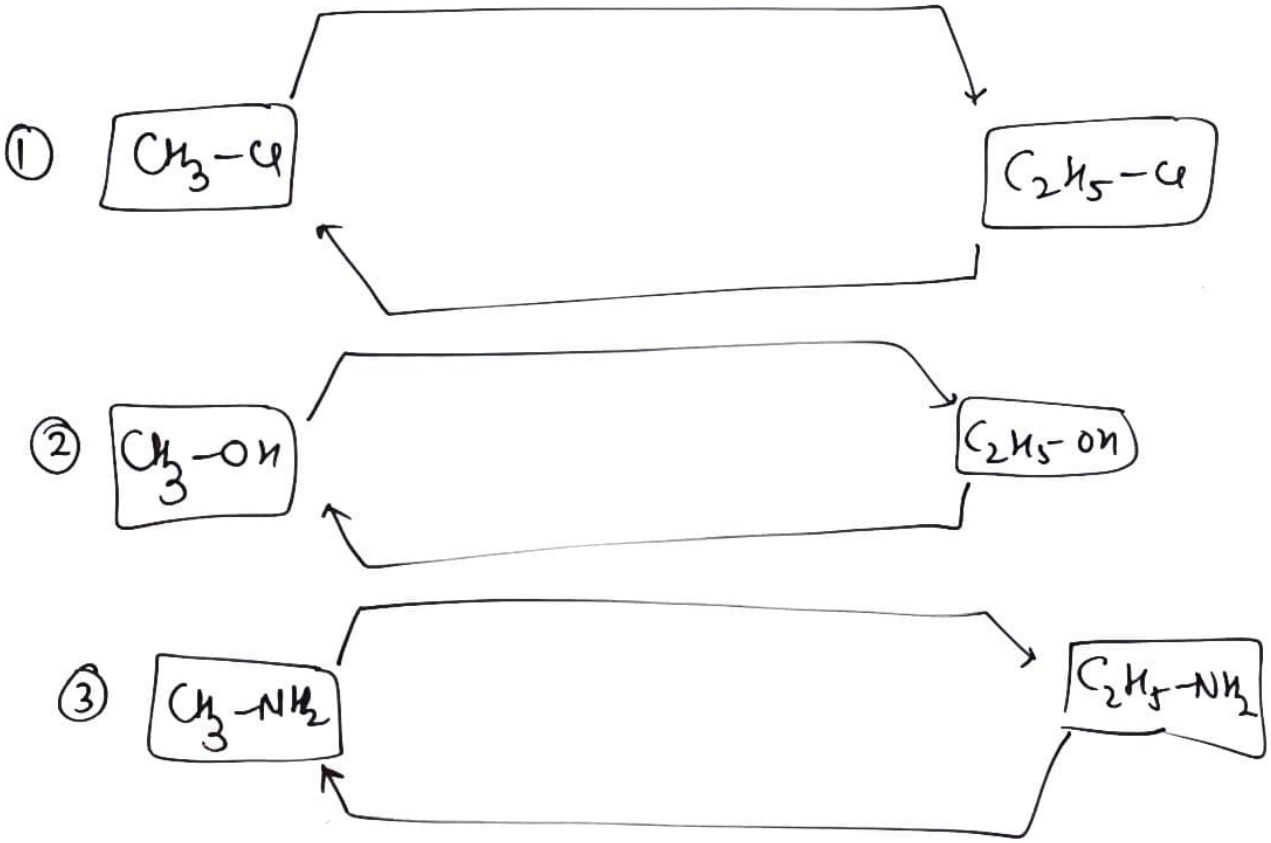
(11.B) Nitration
(11.C) Sulphonation
(11.D) Halogenation.

P > m > o
51% 47% 2%

... in ORM = II

AMINES

Ques. Give Suitable steps for following interconversions.



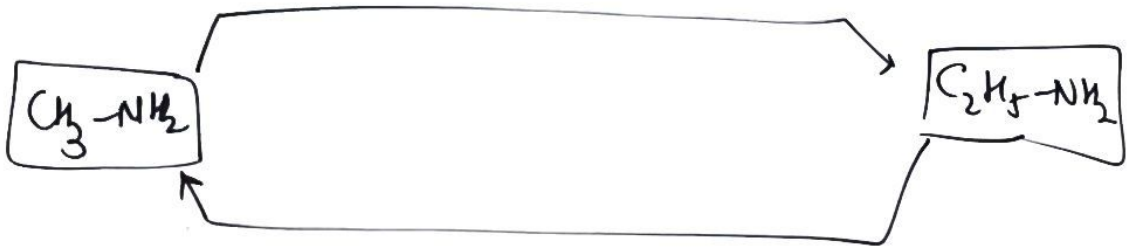
AMINES

Ques. Give Suitable Steps for following interconversions.

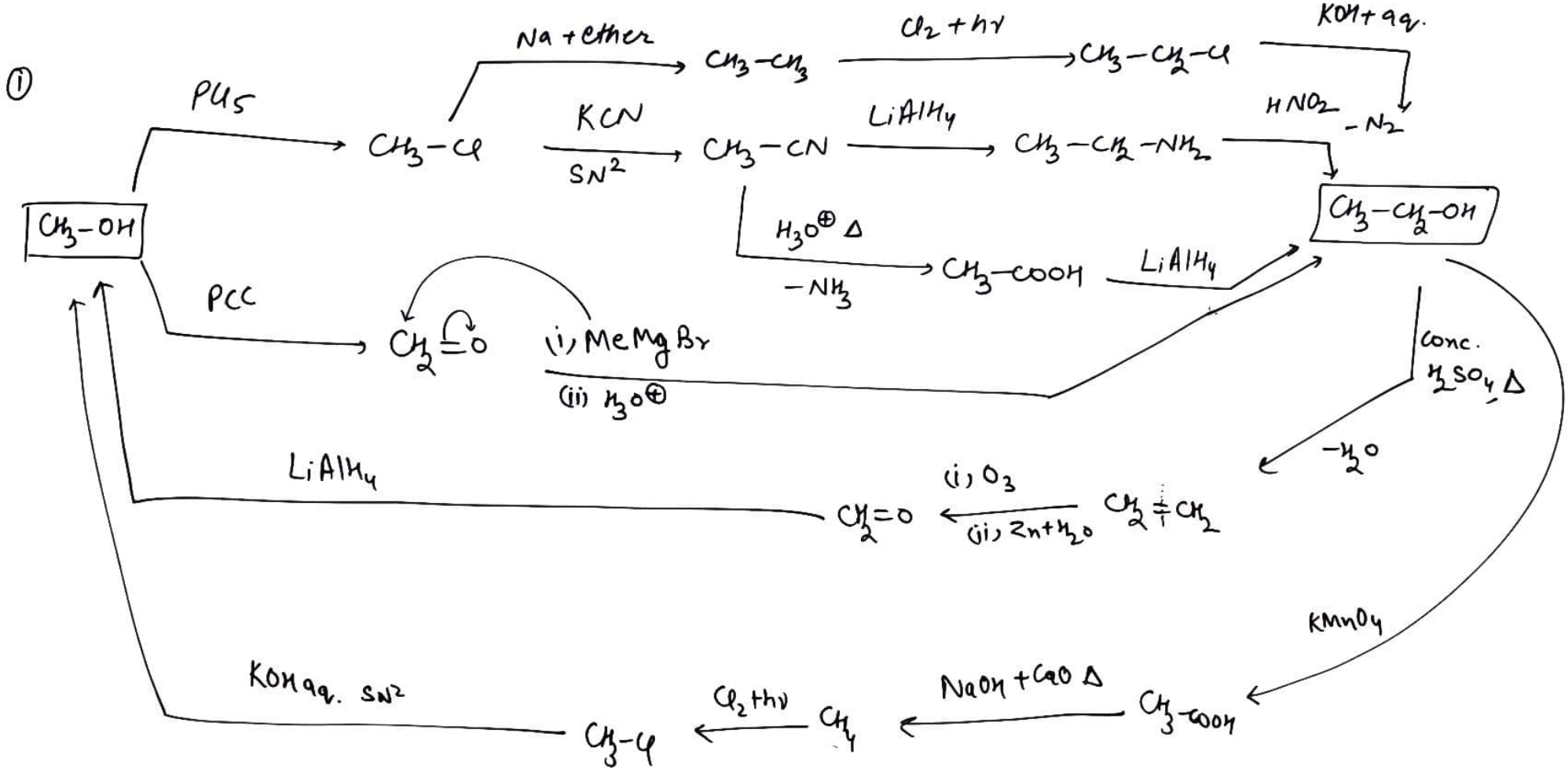
④



③

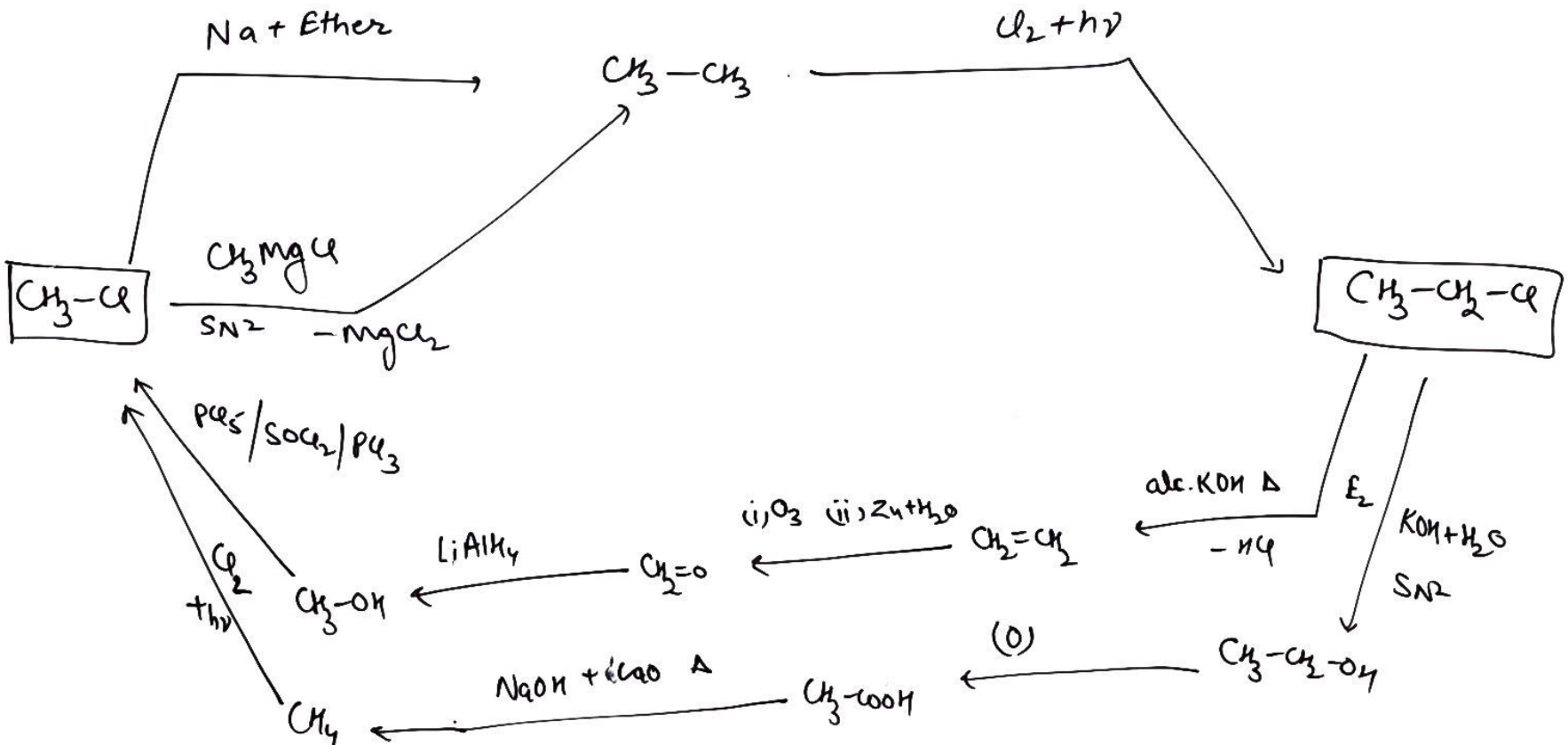


Ques :- Give suitable steps of interconversion...

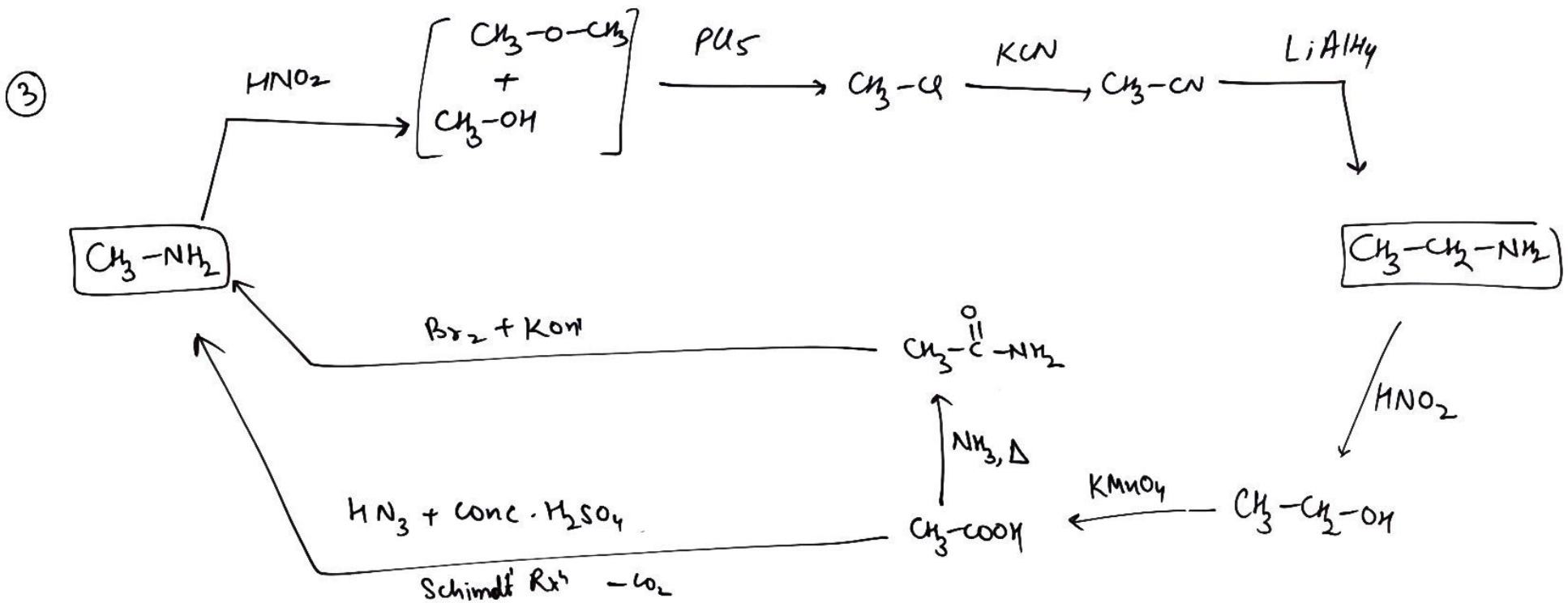


Ques :- Give suitable steps of interconversion...

(2)



Ques :- Give suitable steps of interconversion...



Ques :- Give suitable steps of interconversion...

(4)

